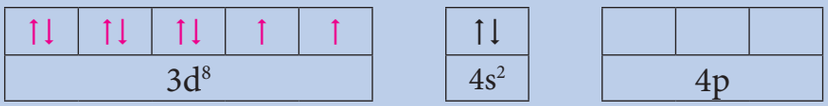
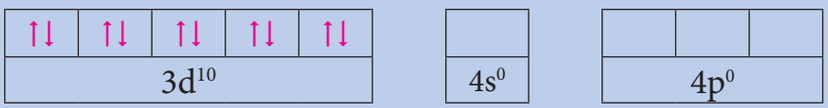
9. Ligands such as CO, CN^-, en, and NH_3 present in the complexes cause pairing of electrons present in the central metal atom. Such ligands are called strong field ligands.

10. Greater the overlapping between the ligand orbitals and the hybridised metal orbital, greater is the bond strength.

Let us illustrate the VBT by considering the following examples.

Illustration 1

Complex	**$[Ni(CO)_4]$**
Central metal atom and its outer electronic configuration	Ni: $3d^8$, $4s^2$
Outer orbitals of metal atom/ion	
Nature of ligand	CO Strong field ligand causes the pairing of 4s electron with 3d electrons in the metal
Outer orbitals of metal atom/ion in presence of ligand	
Hybridisation	Coordination number - 4 Hybridsation - sp^3
Hybridised orbitals of the metal atom in the complex	
Geometry	Tetrahedral
Magnetic property	No. of unparied electrons = 0; Hence diamagnetic
Magnetic moment (Using spin only formula)	$\mu_s = \sqrt{n(n+2)} = 0$

XII U5 Coordination jagan.indd 149 04-01-2021 13:24:15